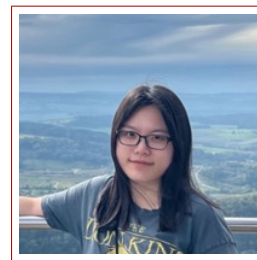


Nina Weng

Curriculum Vitae

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I am a third-year PhD student at the Technical University of Denmark (DTU), working on reliability, fairness, and trustworthiness of machine learning systems in medical imaging. My research spans bias assessment, counterfactual generation, shortcut detection, and bias mitigation, and includes experience with probabilistic and generative models such as diffusion models and explainable methods. My goal is to develop scalable, transparent, and trustworthy machine learning methodologies.

Education

- 2023.3–2026.2 **PhD, Computer Science**, *Technical University of Denmark*, Denmark.
Responsible AI for medical imaging (Algorithmic Fairness, Explainability, Counterfactuals, Bias reasoning, Generative Modeling)
Supervisors: Aasa Feragen (principle), Siavash Bigdeli (co-supervisor)
Expected completion date: 28th Feb, 2026
- 2025 Spring **Visiting PhD Student**, *University of Luzern*, Switzerland.
Supervisor: Christian F. Baumgartner
Working on subgroup discovery for segmentation tasks.
- 2020–2022 : **M.Sc. in Human-centered Artificial Intelligence**, *Technical University of Denmark*, Denmark.
GPA: 10.47/12, Deep Learning, Explainability, Time series data
- 2021–2022 : **Exchange Master student**, *ETH Zürich*, Switzerland.
Focus on Deep Learning & Computational Neuroscience
- 2016–2020 : **B.Eng. in Computer Science and Technology**, *Beihang University*, China.
Focus on Deep Learning & Computer Vision
- 2019 Spring: **Exchange Bachelor student**, *Politecnico di Torino*, Italy.

Awards & Achievements

- 2025 Travel grant from *Otto Mønsted Foundation* (20,000 dkk, $\approx$ 2600 Euro)
- 2025 Travel grant from DTU for external stay (25,000 dkk, $\approx$ 3300 Euro)
- 2025 Nomination from DTU for Elite Research travel grants
- 2024 *Otto Mønsted Foundation's* grant for attending conference abroad (7500dkk, $\approx$ 1000 Euro)
- 2024 ECCV — Oral Presentation (top 2.3% of accepted papers)
- 2023 Departmental Scholarship for PhD study from DTU Compute
- 2022 Graduated as an Honours M.Sc. student from Honours Programme DTU (top 5%)
- 2021 Semester scholarships from Swiss-European Mobility Programme
- 2020 Tuition fee waiver for 2-year's master study, DTU

Invited Talks

- 2024,2025 **Diffusion Model Meets XAI: Counterfactual Generation for Model Debugging**, ①
Visual Intelligence Seminar series at Norwegian Center for Research Based Innovation; ②
Workshop In the picture: Medical Imaging Datasets.
- 2023 **What causes bias? A case study in detecting the sources of gender bias in Chest X-ray automatic diagnosis**, *CX Collaboratory meetup at Danish Pioneer Centre for AI*

Teaching & Student Supervision

Teaching Assistantship

- 2024-2025 **02517 Responsible AI**, *Technical University of Denmark*, BSc, MSc.
- 2023 **02456 Deep Learning**, *Technical University of Denmark*, BSc, MSc.

Student Supervision

- 2024 **Bachelor thesis: Influence Functions for Label Noise**, *F. Jørgensen.*
- 2024 **Bachelor thesis: Investigating Realism Score**, *G. Nielsen.*
- 2023 **Master's level special course: Automatic Label Denoising**, *K. B. Ulbæk.*
- 2023 **Bachelor thesis: Gender Bias in Big Language Models**, *S. Sterlie*, work published as a workshop paper ECCV.
- 2023 **Master's level special course: Hidden Bias in Chest X-ray**, *V. Olesen*, work published as a workshop paper MICCAI.

Responsibilities

Reviewer

- 2024-present Top tier conferences, e.g. ICLR, ICCV, ECCV, MICCAI, MIDL, workshops @MICCAI/ECCV/CVPR

Academic event organizer

- 2025 **Medical Imaging meets EurIPS (MedEurIPS) workshop @ EurIPS**, *Organizing Team.*
- 2024-2025 **Talk series at the Center for Basic Machine Learning in Life Science (MLLS)**, *Talk planning team.*

Society

- 2025-present **High5Girls**, *Role model & Event organizer*, Team member at Personal data courses.
- 2025-present **RISE-MICCAI**, *Storytelling coordinator (interviews & production)*, Youtube@RISEMICCAI, Spotify@Voices of RISE MICCAI.
- 2024-present **Special Educational Assistance (SPS)**, *Academic support teacher.*
- 2020-present **Long-Legs Mailbox Projects**, *Volunteer*, Supporting left-behind children (whose parents work far away from home) in rural China by writing letters to discuss their studies and life.
- 2020-2024 **Women in Tech**, *Editor and Designer*, Contributed to a Douban-born initiative that grew into a non-profit producing blogs, podcasts, articles, and interviews.
- 2020-2021 **DTU student ambassador**, Providing insights of DTU life to prospective students.
- 2018-2019 **A Class in Childhood**, *Teacher*, Supporting children in primary schools in rural areas of China by teaching online lessons throughout the semester. Course taught: Chinese literature.

Publications

Summary of last 3 years, 136 citations, *h-index*:7, *i10-index*:6 (Google scholar), 4 first-author (2 MICCAI, 1 ECCV oral, 1 workshop); 2 workshop papers from student supervision.

- [11] (Pre-print) **Nina Weng**, Aasa Feragen, and Siavash Bigdeli. Patronus: Bringing transparency to diffusion models with prototypes. *arXiv preprint arXiv:2503.22782*, 2025.
- [10] Paraskevas Pegios, Manxi Lin, **Nina Weng**, Morten Bo Søndergaard Svendsen, Zahra Bashir, Siavash Bigdeli, Anders Nymark Christensen, Martin Tolsgaard, and Aasa Feragen. Diffusion-based iterative counterfactual explanations for fetal ultrasound image quality assessment. In *International Workshop on Advances in Simplifying Medical Ultrasound*, pages 174–184. Springer, 2025.
- [9] **Nina Weng***, Paraskevas Pegios*, Eike Petersen, Aasa Feragen, and Siavash Bigdeli. Fast diffusion-based counterfactuals for shortcut removal and generation. In *European Conference on Computer Vision*, pages 338–357. Springer, 2024.
- [8] Sara Sterlie, **Nina Weng**, and Aasa Feragen. Generalizing fairness to generative language models via reformulation of non-discrimination criteria. In *European Conference on Computer Vision*, pages 18–34. Springer, 2024.
- [7] Vincent Olesen, **Nina Weng**, Aasa Feragen, and Eike Petersen. Slicing through bias: explaining performance gaps in medical image analysis using slice discovery methods. In *MICCAI Workshop on Fairness of AI in Medical Imaging*, pages 3–13. Springer, 2024.
- [6] Manxi Lin*, **Nina Weng***, Kamil Mikolaj, Zahra Bashir, Morten BS Svendsen, Martin G Tolsgaard, Anders N Christensen, and Aasa Feragen. Shortcut learning in medical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 623–633. Springer, 2024.
- [5] **Nina Weng**, Martyna Plomecka, Manuel Kaufmann, Ard Kastrati, Roger Wattenhofer, and Nicolas Langer. An interpretable and attention-based method for gaze estimation using electroencephalography. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 734–743. Springer, 2023.
- [4] **Nina Weng**, Siavash Bigdeli, Eike Petersen, and Aasa Feragen. Are sex-based physiological differences the cause of gender bias for chest x-ray diagnosis? In *Workshop on Clinical Image-Based Procedures*, pages 142–152. Springer, 2023.
- [3] Jiahao Wang, Yunhong Wang, **Nina Weng**, Tianrui Chai, Annan Li, Faxi Zhang, and Sansi Yu. Will you ever become popular? learning to predict virality of dance clips. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, volume 18, pages 1–24. ACM New York, NY, 2022.
- [2] **Nina Weng**, Jiahao Wang, Annan Li, and Yunhong Wang. Two-stream temporal convolutional network for dynamic facial attractiveness prediction. In *2020 25th International Conference on Pattern Recognition (ICPR)*, pages 10026–10033. IEEE, 2021.
- [1] Isabel* Barradas, Agnieszka* Kloc, **Nina Weng***, and Jan Treur. A second-order adaptive network model for exam-related anxiety regulation. In *Biologically Inspired Cognitive Architectures Meeting*, pages 42–53. Springer, 2021.

Miscellaneous

Coding	Python, MATLAB, C Language, R, JAVA, PyTorch/TensorFlow, Git, Linux, Latex
Languages	English (Proficient), Chinese: Mandarin/Hokkien (Native), Danish (Intermediate), Spanish (basic), Japanese (basic)
Spare time	Orienteering, Trail-running, Indoor bouldering, Crossfit, Hiking, Writing, Science-fiction, Detective stories